Introduction & Elementary Structures

CSE 4303: Data Structures
Lecture 01

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Lecture Overview

1. Introduction
2. Classification
3. Review of Arrays, Records, Pointers
4. Memory Allocations
5. End

🤔 What is a Data Structure?

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Simply values or sets of values, raw facts or figure without any specific meaning.

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It's about two things:

1. How to store data (structure)
2. How to perform operations on it (efficiency)

💡 The "What" vs. The "How"

The "What": Abstract Data Type (ADT)

An ADT is a mathematical model that defines a set of data and the operations on that data, without specifying how they are implemented.

- 👉 What operations are performed, not how they are performed.
- 👉 Operation on Stack –
  - `push()`
  - `pop()`

The "How":Data Structure (DS)

A DS is a concrete implementation of data storage that allows us to efficiently perform the operations defined by an ADT.

- 👉 How data is organized and stored in memory.
- 👉 Stack implementation –
  - Array →fixed size, random access
  - Linked List →dynamic size,

🤔 Why Study Data Structure?

Applications of Data Structure

- 👉 Computer file system (Data structure maps file names onto hard drive sectors)
- 👉 Google and other search engines (Data structure maps keywords on web pages containing those keywords)
- 👉 What is the longest common subsequence of two DNA can be found?
- 👉 Geographic systems (Data structure find data relevant to the current view/location)
- 👉 Block chain (Linked list)
- 👉 Google Map (Finding shortest distances in terms of distance and time)
- 👉 Data Compression (Huffman's encoding)

Many problems are solved efficiently just using the right data structure. This course is about learning to choose the right data structure for the job.

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- Reading: accessing at a particular element within the data structure.
- Searching: looking for a particular value within a data structure.
- Inserting: adding a new value to the data structure.
- Deleting: removing a value from the data structure.
- Sorting: rearranging element in some logical order.

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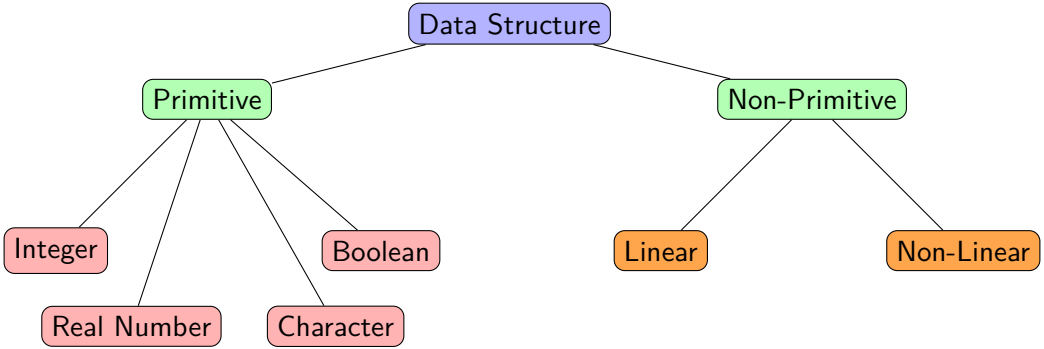
👉 The efficiency — how efficiently allowed operations can be done.
Searching an element in:

- An **unsorted array** takes $O(n)$ time.
- A **balanced binary search tree** takes $O(\log n)$ time.

Classification of Data Structure



Non-Premive Example:

Linear: Arrays, Linked List, Stack, Queue

Non-Linear: Graph, Trees

🧠 Review of Arrays, Records, Pointers I

Arrays:

A collection of elements of the same type stored in contiguous memory locations.

Characteristics:

- 👉 Fixed size defined at declaration
- 👉 Elements accessed using index (starting from 0)
- 👉 All elements are of the same type

Advantages:

- 👉 Fast access via index
- 👉 Simple memory allocation

Disadvantages:

- 👉 Fixed size, cannot grow dynamically
- 👉 Insertion/deletion in middle is costly

🧠 Review of Arrays, Records, Pointers II

Records (Structures):

A collection of related data items of different types grouped under a single name.

Characteristics:

- 👉 Can store heterogeneous data
- 👉 Useful for modeling real-world entities

Advantages:

- 👉 Groups related data together
- 👉 Improves program readability and organization

Disadvantages:

- 👉 Access requires dot operator
- 👉 Not suitable for very large collections without arrays of records

🧠 Review of Arrays, Records, Pointers III

Pointers:

A variable that stores the memory address of another variable.

Characteristics:

- 👉 Enables dynamic memory allocation
- 👉 Used in linked data structures like linked lists, trees, and graphs

Advantages:

- 👉 Efficient memory management
- 👉 Supports dynamic data structures
- 👉 Can pass large structures to functions without copying

Disadvantages:

- 👉 Risk of memory leaks if not managed properly
- 👉 Pointer arithmetic can be unsafe

Memory Allocations I

Memory allocation can be classified into followings:

- Contiguous memory allocation

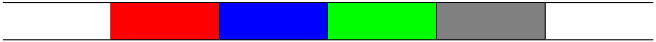


Figure: Contiguous Memory Block (array)

- Linked memory allocation

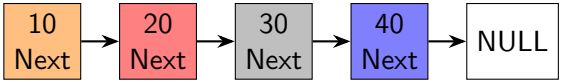


Figure: Linked Memory (Linked List)

Memory Allocations II

- Indexed memory allocation

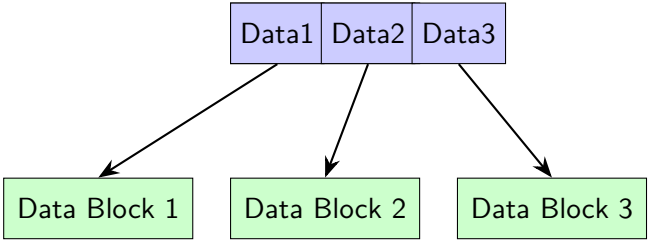


Figure: Indexed memory allocation (Array of pointers)

🚃 Contiguous Memory Allocation

An array stores n objects in a single contiguous space of memory.

- 👉 Can directly access any point randomly. Random access is possible.
- 👉 Unfortunately, if more memory is required, a request for new memory usually requires copying all information into the new memory.
- 👉 In general, you cannot request for the operating system to allocate to you the next n memory locations

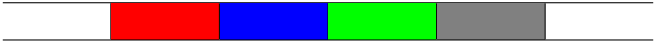


Figure: Contiguous Memory Block (array)

Linked Memory Allocation

Linked storage such as a linked list associates two pieces of data with each item being stored:

- The object itself, and
- A reference to the next item
- Random access to any data apart from the beginning is not possible since the address of a particular data is only stored to its previous data.

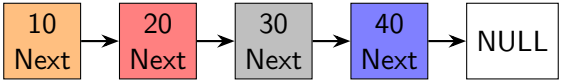


Figure: Linked Memory (Linked List)

Indexed Memory Allocation

An array of pointers (possibly NULL) link to a sequence of allocated memory locations.

Pros:

- No external fragmentation: Data blocks don't need to be contiguous
- Direct access possible: Accessing the index block, you can directly access any block without traversing other blocks.
- Scalability: Large files can grow easily; add more data blocks and update the index block.

Cons:

- Extra memory for index block: Each file needs a separate index block.
- Slight overhead: Accessing a data block requires first reading the index block.

Example: File storing in disk/drives.

🤝 Conclusion & Q&A

Thank You!

👋 Thank you for your time and attention. 👋

🗨 Any questions or clarifications. 💭